



Effect of Scarfing on Rectangular Nozzle Supersonic Jet Plume Flow Characteristics

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An experimental and computational fluid dynamics study is reported of supersonic jets issuing from a high-aspect-ratio rectangular convergent-divergent nozzle with and without a scarfed exit. Schlieren visualization and laser Doppler anemometry measurements captured near-field aerodynamic development of an unheated jet at overexpanded, design, and underexpanded conditions. Reynolds-averaged Navier-Stokes computational fluid dynamics predictions using an eddy viscosity closure (Spalart-Allmaras model) for clean and scarfed geometries were compared with measurements to examine the ability to capture nozzle scarfing effects. The measured plume shape for a scarfed nozzle was strongly affected at overexpanded conditions (a distorted four-lobe shape was observed), whereas a rectangular shape was retained for underexpanded flow although plume bifurcation occurred. The development of the plume shape and the mixing rate was a consequence of the strong vortices that occur with rectangular nozzles, with extra vortices introduced by scarfing. The nozzle exit static pressure changed dramatically when scarfing was added, influencing plume secondary flows and near-field development. The main features of scarfed jet development were predicted qualitatively correctly; the four-lobe overexpanded shape was reproduced but the strength of pressure-driven secondary velocities was overpredicted. The experimental data provided represent a challenging validation test case for computational fluid dynamics studies of three-dimensional supersonic jet plumes with scarfed interaction effects.

Nomenclature

$\mathcal{R}$	= nozzle aspect ratio (at throat, equal to W_t/H_t)
D_h	= nozzle throat hydraulic diameter (equal to $[4W_t H_t]/[2(W_t + H_t)]$)
E	= total energy
H	= total enthalpy
H_t	= throat height (minor-axis dimension)
h''	= fluctuating enthalpy
M	= Mach number
p	= static pressure
q_j	= heat flux
R	= gas constant
U	= time-mean x -velocity component
U_i	= Cartesian velocity component
U_{ref}	= reference velocity (centerline mean axial velocity at nozzle inlet at a given nozzle pressure ratio)
U^*	= nondimensional time-mean axial velocity (x direction, streamwise axis: U/U_{ref})
u_{rms}	= root-mean-square fluctuating axial velocity
u_i''	= mass-weighted fluctuating velocity
u^*	= nondimensional root-mean-square fluctuating axial velocity ($u_{\text{rms}}/U_{\text{ref}}$)
W	= time-mean transverse (z direction) velocity component
W_t	= throat width (major-axis dimension)
W^*	= nondimensional transverse time-mean velocity (z direction, nozzle minor axis: W/U_{ref})
w_{rms}	= root-mean-square fluctuating transverse velocity
w^*	= nondimensional root-mean-square fluctuating transverse velocity ($w_{\text{rms}}/U_{\text{ref}}$)

x	= axial (streamwise) direction
x_i	= Cartesian coordinates
y	= spanwise (horizontal or major axis) direction
z	= transverse (vertical or minor axis) direction
γ	= specific heat ratio
ρ	= density
τ_{ij}	= viscous stress tensor
$\bar{}$	= Reynolds time average
$\sim$	= Favre time average

I. Introduction

IMPROVED propulsion exhaust nozzle design is of increasing importance, with engineering applications such as low noise for civil aircraft and low observability for military air vehicles. In the 1970s, nonaxisymmetric nozzles were identified as offering advantages, including lower cruise/maneuvering drag and enhanced plume/ambient mixing (Hiley et al. [1]). Shyne [2] highlighted the aerodynamic challenges of novel exhaust nozzle technologies, and Gamble et al. [3] reviewed current approaches to nozzle selection/design. For military aircraft, survivability demands low-observable properties. Reduced radar cross section (RCS) requires modification to intake/exhaust geometries, increased wing sweep, and surface shaping. Low infrared signature (IRS) requires cooling of hot surfaces and rapid jet plume temperature reduction. The link between enhanced plume mixing and nozzle shape has been well documented; early NASA studies (Lamb et al. [4]) described the benefits of rectangular nozzles for both RCS and IRS, and Mahulikar et al. [5] provided a review of nozzle geometry options for IRS suppression. High-aspect-ratio rectangular nozzles have also achieved reduced noise for civil aircraft applications (Bridges [6] and Zaman [7]). Nozzle trailing-edge cutouts (notches or slots) further enhance plume mixing by introduction of streamwise vorticity into the jet near field. Experimental measurements demonstrating IRS reduction were provided by Dix et al. [8] for a convergent round nozzle; the hottest part of the plume was shortened by $\sim 33\%$ with little thrust loss. This technology (chevrons or serrations) has, however, been used more for jet noise than IRS reduction.

Integration of the exhaust system and aircraft afterbody combined with low RCS shaping leads to change of the nozzle exit plane angle

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with respect to the plume flow direction. In the F117, for example (Brown [9]), a “letterbox” nozzle with an exit plane parallel to the wing trailing-edge sweep created an angled but still basically rectangular geometry. A nozzle exit plane that is nonorthogonal to the nozzle axis is typically described as “scarfing.” Particularly at offdesign conditions, this can contribute significantly to jet near-field mixing and cross-sectional shape change. The focus of the work reported here is placed on this aspect, i.e., a study of nozzle scarfing effects on near-field plume characteristics for high-aspect-ratio rectangular nozzles. Operating conditions covering nozzle pressure ratios (NPRs, which are the jet total pressure/ambient static pressure) at over-, fully, and underexpanded conditions are considered. Although hot jets occur in practice [nozzle temperature ratio (NTR) = jet total temperature/ambient static temperature greater than one], the present work only considers unheated jets. A detailed study of round convergent nozzles at high subsonic NPRs by Bridges and Wernet [10] has shown that, when scaled appropriately, their near-field mean velocity and turbulence measurements for hot jets collapsed onto the $NTR = 1$ data. Jet heating reduces the potential core length and near-field jet spreading rate; but, if these are taken into account via axial/radial rescaling, little influence of the hot jet conditions can be seen. It is likely that similar behavior may be expected for rectangular jets, and thus the present work concentrates on gathering aerodynamic data at unheated conditions, leaving potential core/spread rate effects of jet heating for rectangular jets to be established in a future study.

The complex fluid mechanics of rectangular jets has been studied for 40 years. Behrouzi and McGuirk [11] summarized published literature on the near-field aerodynamics at both subsonic and supersonic conditions. Two (connected) features appear that are not present in axisymmetric jets: “saddle-shaped” velocity profiles and “axis switching” (the exchange of orientation of the major/minor axes of the jet from the nozzle exit configuration). These features are introduced by the secondary flows (in the plane of the jet cross section) generated in three-dimensional (3-D) jets. Hussain and Husain [12] were the first to explain how near-field entrainment and the developing jet cross-sectional shape for nonaxisymmetric jets are controlled by the azimuthal turbulent structures created at the nozzle lip. These roll up into vortex rings initially aligned with the nozzle exit geometry. Variations in vortex strength and the radius of curvature around the ring then lead to ring deformation due to Biot–Savart law-induced velocity around the ring. Segments at either end of the major axis have higher induced velocity than segments at minor-axis ends, causing fluid at the major-axis ends to convect faster downstream and radially inward relative to fluid at the minor-axis ends. This out-of-plane distortion is accompanied by rapid entrainment/growth in the minor-axis direction, whereas the major-axis jet edge hardly grows or even moves inward. If this mechanism is vigorous enough, axis switching occurs. The strength of the process is controlled by the (Reynolds number dependent) boundary-layer thickness at the nozzle exit. Thus, it does not always occur and is observed most often at subsonic Mach numbers and at lower Reynolds numbers. Putnam and Mercer [13] and Von Glahn [14], however, noted that the same flow characteristics can also appear at high subsonic and supersonic Mach numbers, and both saddle-shaped profiles and axis switching have been observed in shock-containing supersonic jet flow from rectangular nozzles by Veltin and McLaughlin [15] and Behrouzi and McGuirk [11].

Most of the aforementioned measurements were focused on revealing the flow physics of rectangular jets and were conducted using pneumatic probe instrumentation. For computational fluid dynamics (CFD) validation purposes, turbulence measurements are also needed. Laser-based instrumentation was applied to a rectangular jet plume by Behrouzi and McGuirk [11] for a convergent rectangular nozzle (aspect ratio = 7.35); detailed mapping of the mean velocity and turbulence statistics in major- and minor-axis planes was provided for NPRs corresponding to choked, mild, and strong underexpansions. Although this study is a good example of the level of detail important for CFD validation, it is not entirely satisfactory for the present purpose: it focuses principally on the effect of an addition of an aftdeck on near-field plume characteristics, and the use of a convergent-only

nozzle is not representative of military applications where convergent–divergent nozzles are used and both over- and underexpansion states can occur.

As well as vortex ring deformation, for shock-containing jets, a further phenomenon must be considered: the possible occurrence and influence of screech. Screech is tonal noise produced as a result of an acoustically driven self-excited feedback loop. Shear-layer disturbances (instabilities or large-scale turbulent eddies) grow as they convect downstream of the nozzle lip and interact with shock cells, creating sound waves. These propagate upstream in the subsonic regions at the edge and outside of the jet, and they are reflected/scattered upon reaching the nozzle lip to create new shear-layer disturbances to continue the cycle, with the appearance of “screech tones” in the radiated noise. Observations of screech in imperfectly expanded rectangular jets have been reported by Behrouzi and McGuirk [11], Zaman [16], Krothapalli et al. [17], Alkislar et al. [18], and Valentich et al. [19]. In addition, Raman [20,21] provided a comprehensive review of work on screech in rectangular jets from convergent–divergent nozzles, including the effect of nonuniform nozzle exit geometry. Analysis of the available experimental data clearly showed that screech characteristics (mode, amplitude) were strongly dependent on the shock-cell structure. For screech to occur required 1) the instability to grow to a sufficient size and strength, and 2) the pressure change across the shock cells to be sufficiently large. Thus, for a straight rectangular nozzle exit, the source of the screech tone is often considered to be located at the third or fourth shock cell. Although the predominant mode for a straight nozzle exit geometry is a transverse antisymmetric (flapping orthogonal to the nozzle major axis) and spanwise symmetric motion, Raman [20] showed that a spanwise antisymmetric mode could also be found when the shock-cell structure was weakened near the nozzle design NPR. Of course for a complex-geometry nozzle trailing edge, strong spanwise variations in the shock cell structure are inevitable, and hence both spanwise symmetric and antisymmetric modes are always observed. Raman [20] studied two types of nonuniform nozzle exit: a spanwise oblique edge (single-beveled design), and a double-beveled arrowhead-shaped edge (referred to here as scarfed). It was observed that, with the scarfed design, the onset of screech was shifted to a higher value of NPR, screech amplitude was typically weaker and over a narrower range of NPRs as compared to straight and single-beveled shapes, and both spanwise symmetric and antisymmetric modes occurred.

The aforementioned studies indicated that, if the intensity of the screech tone was high, interaction with the mean flowfield could occur and the streamwise vortex structures (and consequently the possibility of axis switching) can be strengthened. However, it has also been noted that weak screech does not interact strongly with the mean flowfield: merely superimposing an unsteady periodic component onto the turbulent fluctuations within the potential core region. For example, Krothapalli et al. [17] observed no change of the jet spreading angle from its nonscreech value for NPRs of 2.4 and 5.4, in spite of clear evidence of screech; whereas at the peak screech condition of $NPR = 3.8$, a 60% increase in spreading was observed. Similarly, Behrouzi and McGuirk [11] noted no mean flow effects of screech at a mild underexpanded NPR but significant effects for strong underexpansion when screech intensity increased. In Alkislar et al.’s [18] measurements, high-amplitude screech only occurred at specific NPR values in both over- and underexpanded jets, and just a $\pm 15\%$ deviation from the peak noise NPR led to a sound intensity reduction of an order of magnitude. Finally, it must be borne in mind that, although the screech phenomenon is of significant interest fluid mechanically, the structurally damaging effects of high-intensity screech mean that, in practical applications, design efforts are undertaken to avoid the offdesign NPR operating conditions that involve strong screech tones (heated jets also are less prone to screech). Thus, when considering test cases suitable for assessing the capability of CFD for design, it is necessary to avoid high screech intensity conditions, and this approach has been followed in the present work.

The presence of nozzle scarfing undoubtedly introduces extra effects. Lilley [22] described design methods for scarfed nozzles in missiles where flush-mounted exit through the side rather than the

rear of the vehicle was the first occurrence of scarfing. Unfortunately, only internal expansion processes with no application to external plume development were considered. Some research has been undertaken to understand what effect scarfing has on vortex ring dynamics. New [23] and New and Tsovolos [24] performed particle image velocimetry measurements on elliptic nozzles with several scarf angles. For low angles of incline, the vortex ring remained aligned with the original nozzle exit; but, for larger angles, the incline increased downstream. Strong axis switching occurred in the nonscarfed elliptic nozzle, but both 30 and 60 deg inclined nozzles produced significant suppression of the axis-switching behavior. It was argued that these observations were consistent with the vortex ring self-induction process, but caution must be used if these conclusions are to be considered generally valid because all measurements were at low Reynolds numbers Re (~ 2500) and at incompressible (water flow) conditions.

Rather few studies have been published on rectangular nozzle scarfing at high Mach numbers, with little detailed measurements available. Samimy et al. [25] and Kim and Samimy [26] carried out experiments on rectangular nozzles with different trailing-edge angles to examine enhanced mixing and noise reduction benefits, with some attention given to plume development, although via flow visualization only. The baseline nozzle geometry was convergent-divergent with an aspect-ratio-of-three exit shape operated at moderately overexpanded ($M = 1.75$), fully expanded ($M = 2.0$), and moderately underexpanded ($M = 2.5$) conditions. Scarfed and sawtoothed trailing-edge shapes were explored at sweep angles of 35, 45, and 55 deg. Scarfing made little difference to the jet cross-section evolution for the ideally expanded case, with no axis switching and gradual development into an axisymmetric shape. In contrast, pairs of streamwise vortices were generated at the underexpanded condition, the spanwise surface pressure gradient introduced by scarfing being identified as the source. In the overexpanded flow regime, an internal nozzle flow separation prevented a firm conclusion from being drawn. Once again, caution is warranted when generalizing these observations. Finally, as noted previously, Raman [20,21] provided a study of a 4.75:1 aspect ratio scarfed rectangular nozzle at overexpanded, design, and underexpanded conditions. To understand the shock-cell structures produced, a small pneumatic probe was employed to document the static pressure field on axial planes through the major and minor axes of the jet plume. However, no quantitative velocity or turbulence measurements were carried out and the detailed effects of scarfing on the supersonic cores of high-speed rectangular jets remain to be established.

On the basis of the preceding review, it is clear that it would be valuable to capture detailed flowfield measurements in a scarfed rectangular convergent-divergent high-aspect-ratio nozzle jet plume for a range of representative NPR operating conditions. The main objective of the present work was therefore to produce a measurement database of velocity and turbulence properties for both a plain exit plane and a scarfed exit plane. In addition, because no CFD studies to predict scarfing effects have been carried out, a second objective was to use the data taken as a CFD validation test case to establish the ability of a Reynolds-averaged Navier-Stokes (RANS) eddy viscosity turbulence model to predict scarfed rectangular supersonic jets.

II. Experimental Facility and Instrumentation

A. High-Pressure Nozzle Test Facility

Measurements were carried out in the High-Pressure Nozzle Test Facility at Loughborough University. Behrouzi and McGuirk [11] provided a full description of the facility and instrumentation, and only brief details are provided here. A maximum flow rate of 1 kg/s of high-pressure air at a maximum pressure of 14 bar (gauge) and a dew point of -40°C is available, with 110 m³ of air storage provided for blowdown testing at higher flow rates; automatic control valves hold the mass flow constant to $\pm 1\%$.

B. Rectangular Nozzle Design

Two convergent-divergent nozzles (clean and scarfed) were designed. The clean design had inlet, throat, and nozzle exit heights of 25.0, 13.1, and 15.75 mm, with the width remaining constant at 91.6 mm. The nozzle exit lip thickness was chosen to be as thin as possible, adopting the standard manufacturing practice. This aspect of nozzle geometry will influence the particular screech characteristics of the nozzle, and thin nozzle lips minimize (but do not eliminate) the peak screech amplitude; a 1 mm thickness was deemed acceptable. The throat hydraulic diameter ($D_h = 22.92$ mm) was chosen as the reference length for coordinate nondimensionalization. The convergence-divergence angles and axial lengths were 6 deg /56.6 mm and 1.36 deg /55.78 mm. Ideal expansion was calculated to be at NPR = 4 (fully expanded Mach number = 1.56). This was an inviscid approximation ignoring losses and inlet boundary-layer effects. The actual design NPR was identified using flow visualization. Note also that the throat corner had a rounded shape with a radius on the order of 10 mm; thus, flow-turning-created weak oblique pressure waves occurred inside the nozzle, even at the design NPR, and these could be seen in the jet plume as shown below in Sec. V.A. The scarf angle for the second nozzle was set at 35 deg, which is at the lower end of the range used in previous scarfed nozzle measurements. The divergence length measured on the nozzle centerline was increased to 71.55 mm. The divergent section had an initial unscarfed length and was designed so that the nozzle exit area projected into the axial direction was equal to the clean nozzle exit area; the scarfed nozzle exit then intersected the nozzle sidewall a short distance downstream of the throat. Thus, in a one-dimensional (1-D) design context, the area ratios of both nozzles were the same. Figure 1 presents photos of clean and scarfed nozzles (CN and SN, respectively). The coordinate system (note that the origin is at the throat geometric center) used to report measurements and CFD predictions is also shown. The high-pressure air supply pipeline consisted of a 75 mm round pipe; a round to rectangular transition duct (75 mm diameter to 25 mm high $\times$ 91.6 mm wide) that was 275 mm long was attached to the pipeline and was followed by a straight rectangular duct with a length of 677 mm ($29.5D_h$), to which the nozzles were connected using the flange shown in Fig. 1. Optical access windows were introduced into the side and upper wall of the nozzle supply duct at a distance of 154 mm ($6.73D_h$) upstream of the nozzle throat plane and within the straight rectangular section to allow measurement of the wall boundary layer that had built up in the supply duct. Laser Doppler anemometry (LDA) measurements along the major and minor duct axes showed that the 99% boundary-layer

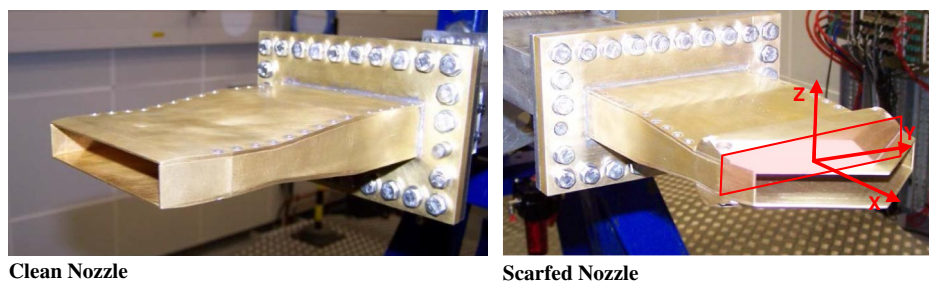


Fig. 1 Views of the clean nozzle (left) and scarfed nozzle (right).

thickness was 5.5 mm on both walls; the mean velocity followed a one-ninth power law velocity profile, and data for the axial Reynolds stress closely matched classical measurements for a zero-pressure gradient boundary layer. The availability of this information allowed this location to be treated as the nozzle inlet plane in the CFD calculations reported in the following.

C. Instrumentation

Schlieren flow visualization and two-component LDA mean velocity and turbulence measurements were conducted throughout the near-field region of the jet plume. The schlieren unit used a rainbow filter instead of a knife edge; undeflected light showed yellow, and deflected light showed red in expansion and green in compression regions. Schlieren photographs were taken of the major- and minor-axis planes by rotating the nozzle. The DANTEC LDA system was used in backscatter mode and contained an aluminium oxide (Al_2O_3) solid particle generator ($0.3\ \mu\text{m}$ median diameter) to seed the flow. The LDA-measuring volume dimensions (Δx , Δy , Δz) were 0.076, 1.254, and 0.076 mm; 20,000 validated samples or a 20 s sampling time (whichever fulfilled first) were found suitable for the capture of smooth, repeatable statistics. No corrections were made for the sampling bias; rather, associated errors were minimized by using high data rates as compared to velocity fluctuation levels, and samples were gathered at fixed time intervals (data rate of 2 kHz). The measurement uncertainties for the mean velocity/rms values were estimated as $\pm 2\%$ and $\pm 5\%$.

III. Computational Modeling

A. CFD Code, Numerical Details, and Turbulence Model

Prediction of unscarfed round and rectangular jets has already been considered in the published literature: in particular, to establish the accuracy of predicting the potential core length. For axisymmetric jets, Georgiadis and DeBonis [27] presented an extensive review of the RANS capability for potential flow length prediction. The main conclusions were as follows:

- 1) Standard two-equation closures consistently overpredict the potential core length due to inhibited initial turbulence growth rate, and they display too rapid initial centerline decay.

- 2) Modifications to improve near-field jet mixing accuracy had variable results (the modified $k-\epsilon$ model of Thies and Tam [28] did improve core length prediction, but the downstream axial decay was then significantly too slow). Wang and McGuirk [29] have demonstrated that, if careful attention is paid to initial conditions and to subgrid-scale modeling, the large-eddy simulation (LES) approach can predict the potential core length accurately for clean rectangular supersonic jets. The LES is, however, an expensive methodology for routine design. Given that no computational investigation has been carried out on scarfed nozzle jets, the objective of the CFD work here was to assess the capability of a standard RANS closure to predict the changes in flowfield introduced by scarfing, accepting the known deficiency of RANS for potential core length prediction. Only steady RANS solutions were attempted here. As will be described in the following, efforts were undertaken to ensure the detailed measurements were carried out at NPRs that avoided intense screech. If strong screech was present, the interaction between unsteady motions and the mean field meant an unsteady RANS or potentially LES CFD approach would be necessary. The present investigation was considered worthwhile to examine the capabilities of steady RANS and an industry-standard turbulence model for the test conditions selected.

The Rolls–Royce Hydra CFD code (Crumpton et al. [30]) was used in the present work. This is a compressible density-based code that adopts unstructured finite volume discretization (on node-centered median-dual cells). Discretization of the inviscid fluxes corresponds to central differencing plus fourth-order smoothing; monotonic resolution of sharp gradients is obtained by constructing the smoothing operator to include a blend of second- and fourth-order differences and a nonlinear limiter. A multigrid, parallelized method is adopted to ensure high efficiency; the message passing implementation was described by Hills [31]. The equations solved are the standard Favre-averaged Reynolds-averaged Navier–Stokes equations for mass,

momentum, and total enthalpy; auxiliary algebraic equations to calculate the temperature from the total enthalpy (for use in the ideal gas relation to evaluate the pressure) complete the set. Finally, the turbulence model adopted to evaluate the turbulent momentum and enthalpy fluxes was the Spalart and Allmaras (S-A) [32] one-equation eddy viscosity model commonly used in aerospace applications. Preliminary calculations were performed for a clean-nozzle $\text{NPR} = 2.5$ case with both $k-\epsilon$ and S-A models. Both models predicted the same shock cell number with similar amplitudes. For the potential core length, predicted values were $7.2D_h$ (S-A) and $8.6D_h$ ($k-\epsilon$), as compared to a measured value of $\sim 6D_h$; because the S-A model produced a more accurate estimation (although still overpredicted as expected), this model was retained in all further calculations. The high-Reynolds-number form of the model was applied, using a wall-function approach near all solid surfaces.

B. Solution Domain and Mesh

The same solution domain was chosen for both clean and scarfed nozzles; both designs were symmetrical about the z axis, and thus a half-nozzle geometry was chosen for meshing (Fig. 2). A short section of the rectangular inlet air supply duct to the nozzle was included in the geometry to allow measured inlet condition information to be applied; the nozzle inflow plane was thus at $x/D_h = -6.73$.

The spread of the jet due to entrainment was visualized for both nozzles using schlieren imaging for a range of NPRs over a distance of $\sim 20D_h$ downstream of the nozzle exit. This allowed estimation of the size of computational volume external to the plume to allow unconstrained ambient entrainment to occur, but without minimizing the external region mesh required. Following Babu and Manesh [33], a short “buffer” region was included upstream of the nozzle to enable smooth merging of the external entrainment flow induced (effectively, a small “coflow” was created); the inlet plane for the external region was thus placed at the nozzle throat axial location ($x/D_h = 0$), so any upstream entrainment could adjust before encountering the jet. The outer boundary of the external region was fixed at a radius of $6D_h$ from the nozzle centerline, and the downstream outflow exit boundary was at $x/D_h = 22.43$. Figure 2 illustrates the solution domain and nonuniform mesh spacing used to ensure adequate resolution in regions of expected high-velocity gradient: particularly over the scarfed nozzle exit plane. To maximize grid quality, a structured hexahedral mesh with 33 blocks was generated. Optimization of the mesh smoothness between blocks was given special attention, particularly at the nozzle exit plane and over the initial $10D_h$ axial distance. To choose the mesh density, an offdesign (overexpanded) case was considered (to ensure shock cells were present). The mesh number was increased from an initial grid of 1.5×10^6 nodes until 1) the amplitude of velocity oscillations in the jet core; and 2) the jet potential core length, changed by less than 5%, leading to a final mesh size of 4.28×10^6 cells.

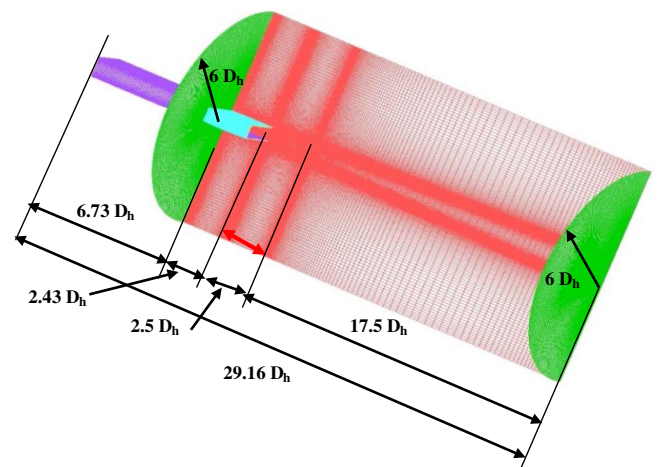


Fig. 2 Computational solution domain and mesh on the $y/D_h = 0$ plane (clean nozzle).

C. Boundary Conditions

At the nozzle inlet plane for each test case NPR, a specified total pressure distribution was applied using the nozzle inlet measurements to include the low total pressure boundary-layer regions, with a uniform total temperature of $T_0 = 288K$; static pressure was extrapolated upstream from the first mesh points inside the solution domain and used to calculate the mean velocity (assuming axial flow). The turbulent eddy viscosity in the Spalart–Allmaras model at this plane was set at $20\mu_{\text{mol}}$. Symmetry conditions were used on the plane of $y/D_h = 0$. Constant static pressure and temperature conditions were applied on the coflow inlet plane, on the radially outermost plane (at $r/D_h = 6$), and on the downstream outlet plane. The static pressure was set equal to ambient pressure on the radial outer and downstream outlet planes, but it was slightly higher (~ 50 Pa) on the coflow inlet plane to provide a small entrainment flow; the turbulent viscosity on the coflow inlet plane was set at $2\mu_{\text{mol}}$.

IV. Test Conditions and Datasets

The selected parameters for the LDA measurement were the mean and turbulent rms velocities in the x and z directions (U , u -rms, W , and w -rms) and the turbulent shear stress $u'w'$. These are presented in nondimensional form ($U^* = U/U_{\text{ref}}$, $u^* = u\text{-rms}/U_{\text{ref}}$, $w^* = w\text{-rms}/U_{\text{ref}}$, and $uw^* = u'w'/U_{\text{ref}}^2$); the reference velocity was the measured centerline velocity at the nozzle inlet plane for each NPR. Schlieren visualization was first used for NPRs of 1.8–5.0 to identify design points for both nozzles. For the clean nozzle, NPR = 3.5 was the most shock-free condition; this reduced to 3.15 for the scarfed nozzle due to different losses and changed expansion conditions introduced by the scarfed geometry. Three operating conditions were then selected for detailed measurements: overexpanded NPR = 2.5, design NPR = 3.5 (clean) and 3.15 (scarfed), and underexpanded NPR = 4.0; the over- and underexpanded NPR values were purposely chosen to avoid the peak screech NPR condition. For the clean nozzle, velocity traverses along y and z axes (35/50 measurement points) were made at 16 axial stations between $x/D_h = 2.93$ and 12.43. A centerline traverse (60 points) up to $x/D_h = 17.93$ was also made. For the scarfed nozzle, four planes were selected for study: three y – z planes (half-planes: $y/D_h < 0.0$) at $x/D_h = 4, 7$, and 12. In addition, a single x – z plane ($y/D_h = 0.0$) was measured, and a total of 79 measurement points covering $1.8 < x/D_h < 24.0$ captured the centerline development (the cutback nozzle exit allowed measurements to start inside the scarfed nozzle).

V. Results and Discussion

A. Schlieren Images

Figure 3 (left) presents five minor-axis plane schlieren images of the clean nozzle flow at NPRs between 2.5 and 5.0, and it presents two major-axis views at over- and underexpanded conditions. The short black lines at the bottom indicate axial spacings of D_h . At NPR = 2.5, an overexpanded pattern appears, with the terminating normal shock at the divergent duct exit just visible and bending out into the plume. At NPR = 3.0, the strength of the compression region decreases and oblique waves appear. A much reduced wave amplitude appears for NPR = 3.5, confirming this is close to the clean nozzle design NPR (which is lower, at probably around 3.4, although no measurements were made to confirm this). Underexpanded oblique waves of increasing strength and axial extent are observed for NPR = 4.0 and 5.0. The change in color of the first wave system from green at NPR = 3.0 to red at 4.0 is further evidence that these lie on either side of the fully expanded condition. The images display a symmetric structure with strong expansion and compression regions, which weaken axially due to imperfect reflection of pressure waves from the thickening shear layer at the jet/ambient boundary. The major-axis view for the overexpanded NPR = 2.5 indicates a strongly two-dimensional (2-D) spanwise uniform wave pattern over the central 80% of the major axis; at the underexpanded condition, a mixed wave pattern appears: a central 2-D system for three cells with an oblique wave pattern superimposed, which originates from the flow turning inside the nozzle and the nozzle

corners. Note the jet width appears to decrease in the major-axis direction at NPR = 5.0, which may be evidence of the start of a mild axis-switching process; the overexpanded major-axis view shows much less width decrease.

Finally, the minor-axis view for NPR = 3.0 indicates a wider and more diffuse outer region of the plume (from $\sim 3D_h$ downstream of the nozzle exit) in comparison to smaller/larger NPRs. This is evidence of transverse jet flapping caused by screech, which was observed to be strongest at NPR = 3.0. Videos of the plume showed the upstream wave structures were caused to oscillate transversely (z direction) and, at this peak screech intensity, the amplitude was large enough that interaction with the overall mean flow behavior was clearly observed. As the NPR increased or decreased away from NPR = 3.0, the screech amplitude decreased rapidly. NPR = 2.5, NPR = 3.5, and NPR = 4.0 were chosen as the overexpanded, (close to) design, and underexpanded conditions at which the screech was weak enough to not affect the mean flow. Measurements (see the following) confirmed that, at these chosen conditions, although effects of weak screech could still be seen in the unsteady motion fluctuating statistics within the potential core, there was no evidence of this in turbulence stress measurements within the jet shear layers that control the mixing of mean momentum.

Figure 3 (right) provides schlieren images for the scarfed nozzle over the same NPR range, indicating a strong effect on shock cell structures. Scarfing causes laterally variable expansion inside the nozzle as the axial length of divergence varies spanwise (y direction). The scarfed nozzle design NPR is lower than for the clean nozzle, as shown by the NPR = 3.15 image, which shows an essentially shock-free flow. All minor-axis shock cell images are less sharp than for the clean nozzle. This is caused by interaction between extra oblique shocks from the angled nozzle lip body (which allows substantially more flow turning) and the line-of-sight lateral averaging of the more 3-D wave structures by the schlieren technique; these changes are mostly associated with the near nozzle exit region. The major-axis view at NPR = 5.0 indicates that all signs of axis switching have now disappeared as the lateral jet edge displays expansion rather than contraction, which is another change induced by scarfing. There were also no indications of strong screech at any NPR for the scarfed nozzle; this is consistent with the observation of Raman [20] that scarfing shifts the onset of screech to higher NPR: in this case, above the highest NPR tested.

B. LDA Measurements

1. Clean Nozzle Plume Development

Figures 4 and 5 present the development of nondimensional time-averaged U^* , u^* , and w^* rms along the nozzle centerline for the clean nozzle at all three NPR conditions (scarfed nozzle results are also shown, and they are discussed in the following section). For the overexpanded case, Fig. 4 (top) indicates that a curved shock exists at the nozzle exit, and thus subsonic velocities are measured just downstream of the exit. The flow then rapidly accelerates and undergoes pressure-wave-induced oscillations, which have significant amplitude (initially $\sim 50\%U_{\text{ref}}$). If the jet is improperly expanded, the presence of oscillations makes potential core length identification difficult. The velocity about which the decaying oscillations are symmetric has here been taken as the core velocity of the fully expanded jet. Using this definition, the potential core length (calculated from nozzle exit, which is located at $x/D_h = 2.43$) for the overexpanded condition is shortest at $4.3D_h$. At the (close to) design NPR (Fig. 4; middle), much reduced oscillations are observed (only $\sim 15\%U_{\text{ref}}$) and the core length increases to $5.5D_h$. Underexpanded flow (Fig. 4; bottom) sees the oscillation amplitude increase again, and the core length grows to $6.5D_h$. At this latter condition, small oscillations are seen downstream of the potential core end after velocity decay has begun. Little difference is observed in the initial centerline velocity decay rate after the end of the potential core between the three NPRs studied.

Centerline clean nozzle turbulence measurements (Fig. 5) show transverse fluctuations w^* are initially larger than axial u^* at all NPRs, particularly at offdesign conditions but also at NPR = 3.5 because this is not quite at the design condition. The dominant strain rates occur in shear layers and imply that turbulence production is

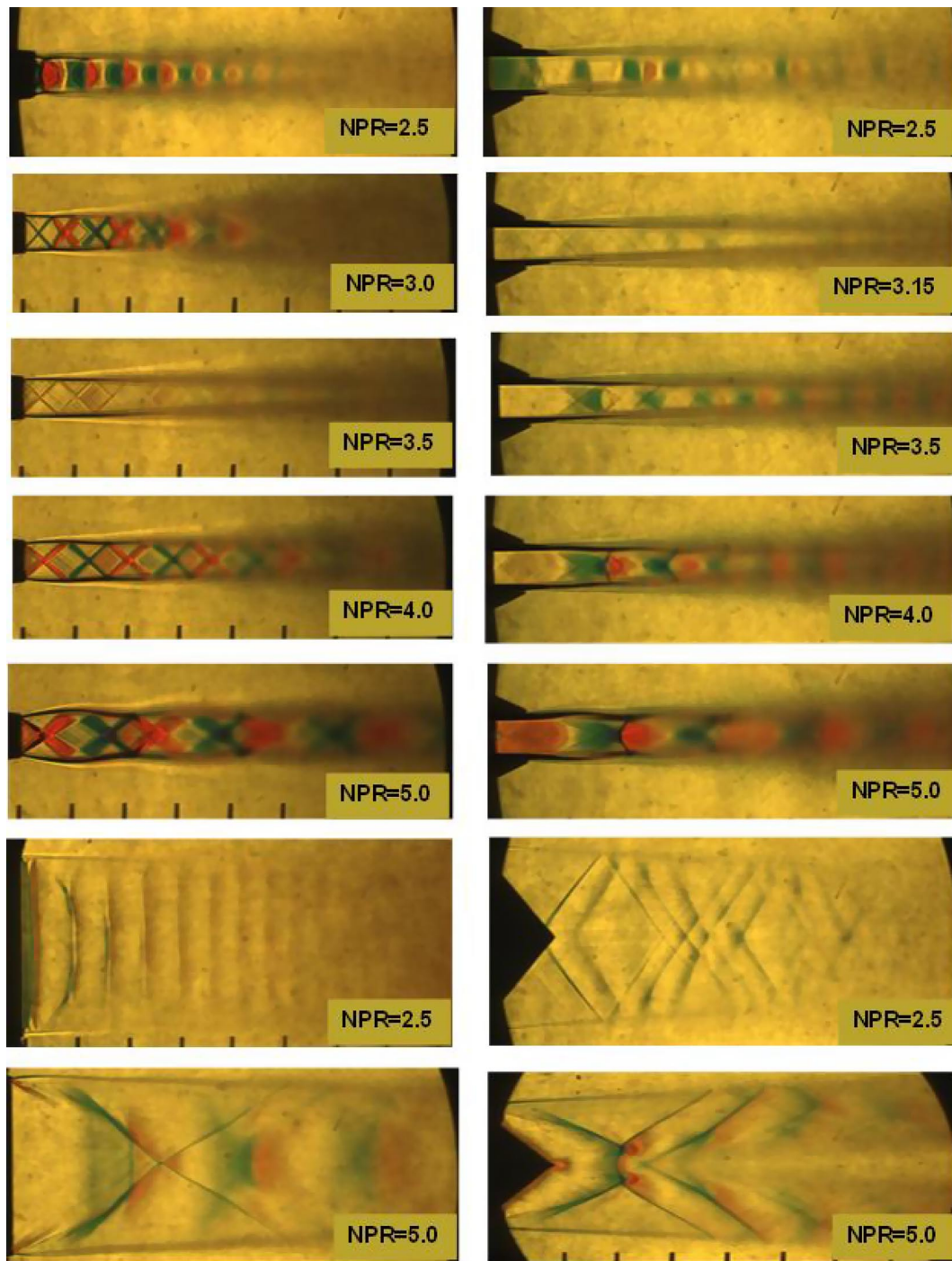


Fig. 3 Side and top (two bottom photographs) schlieren images of clean (left) and scarfed (right) jet plumes.

largest for u^* , which would thus be expected to be greater than w^* . The cause of this centerline anomaly is the effect of the mild screech-induced transverse jet flapping, which adds a periodic unsteady contribution to the transverse turbulence fluctuation. A similar relationship between axial/transverse fluctuations was observed by Alkislar et al. [18] and was confirmed as transverse movement of the oblique shock/expansion wave pattern using phase-locked measurements. The peaks in w^* are coincident with the location of strong pressure-wave interactions. Note that these effects of screech are only visible within the potential core itself; for all NPRs, as soon as the jet shear layers grow to reach the centerline, the expected normal stress relationship of $u^* > w^*$ is seen, confirming that only weak screech effects are present and only within the inviscid core, influencing potential core fluctuation measurements but not affecting important mean momentum mixing regions.

Figures 6 and 7 provide examples of the clean nozzle measured axial velocity and turbulent shear-stress profile data (Fig. 6 also contains CFD results, to be discussed later); in both cases, minor-axis data at three axial stations are provided. Good symmetry is displayed; in Fig. 7, the two zones of high shear stress separated by a reducing zero shear region clearly identify the inward growth of jet/ambient mixing. Note that, in regions where centerline w^* is influenced by jet flapping, the measured shear stress is zero, indicating no correlation between the axial turbulence and transverse flapping motions. Both velocity and stress profiles quantify the monotonic outward spread of the plume in the z direction; the spreading rate is greatest for the overexpanded NPR, and it is approximately the same for the design and underexpanded cases. For the selected locations, only at the underexpanded NPR (4.0) is evidence of centerline compression/expansion wave oscillations visible.

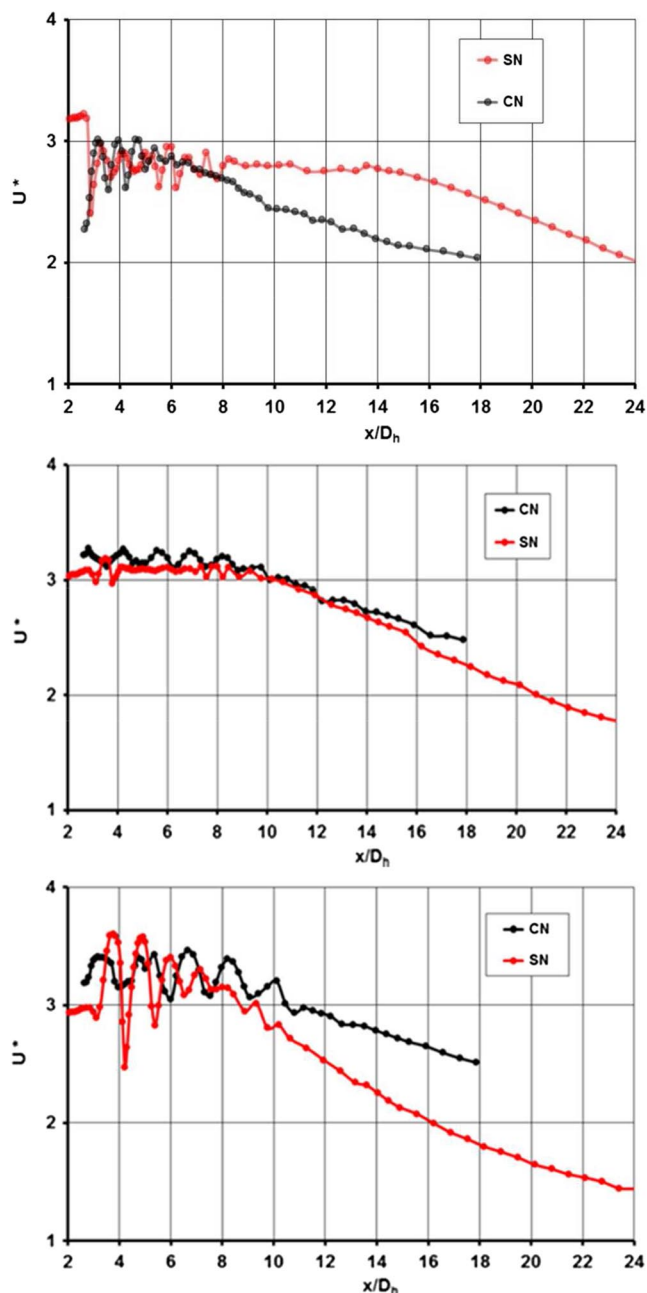


Fig. 4 Mean axial U^* velocity plume centerline, with clean and scarfed nozzle NPRs: 2.5 (top), 3.5 (CN) and 3.15 (SN) (middle), and 4.0 (bottom).

To display flow development in the major-axis y direction, contour plots of U^* and turbulent normal stresses are shown in Figs. 8 and 9. The presence of pressure waves in the jet core is easily identified in the U^* plots. The velocity at the nozzle exit increases with NPR, by $\sim 16\%$ between overexpanded and (close to) design conditions and by a further 3% between (close to) the design and underexpanded NPRs. In addition, the larger downstream extent of the region where the static pressure adjusts to ambient in the underexpanded as compared to the overexpanded flow is plain to see. The more 3-D nature of this process for $\text{NPR} = 4.0$ is illustrated well in Fig. 8, where the islands of higher velocity stretch across the whole inviscid core in the overexpanded flow, but two peaks on either side of the centerline are observed for $\text{NPR} = 4.0$ near $x/D_h = 5$. Further downstream, there is no evidence of axis switching in the axial distance covered by measurements; but, for NPRs 3.5 and 4.0, the appearance of saddle-shaped velocity profiles is seen beyond $x/D_h \sim 9.0$. Because screech effects on the mean velocity development are expected to be small, this is associated with the vortex ring mechanism observed in rectangular jets. Its absence for $\text{NPR} = 2.5$

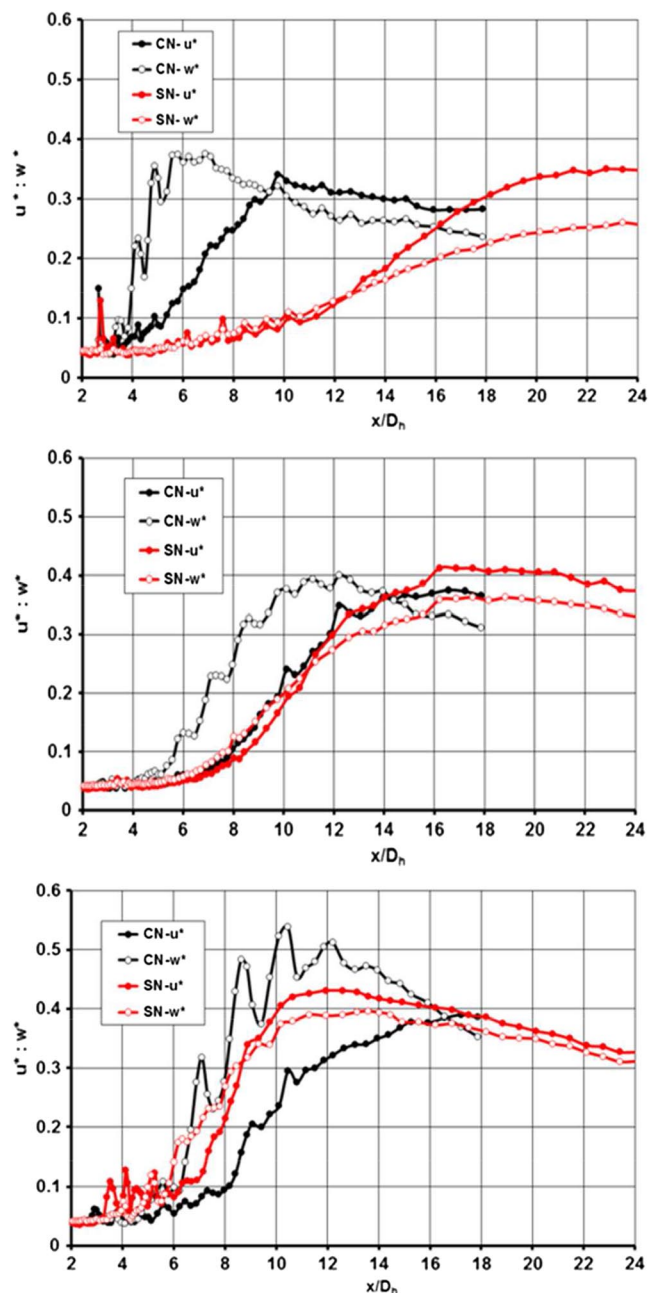


Fig. 5 Axial u^* and transverse w^* rms plume centerline, with clean and scarfed nozzle NPRs: 2.5 (top), 3.5 (CN) and 3.15 (SN) middle, and 4.0 (bottom).

implies that the overexpanded case contains other sources of secondary motions (in the y - z plane), which are contrary to and counteract the vortex ring-driven secondary velocities.

The normal stress contours in Fig. 9 show that, in the major-axis shear layer, the axial stress is always larger than the transverse, as expected from shear-generated turbulence arguments. This indicates that the screech-induced transverse flapping on the centerline within the inviscid core does not influence the jet/ambient momentum exchange process that controls mean flowfield development. Note also in Fig. 9, particularly at $\text{NPR} = 4.0$, the double peak at the furthest downstream locations in the axial stress; this is clearly related to the saddle-shaped axial velocity profile, which creates both outboard (jet/ambient) and inboard shear layers in the mean profile.

2. Effect of Scarfed Nozzle Geometry on Plume Development

First observations on the different development of jets from clean and scarfed nozzles are made using the plume centerline development

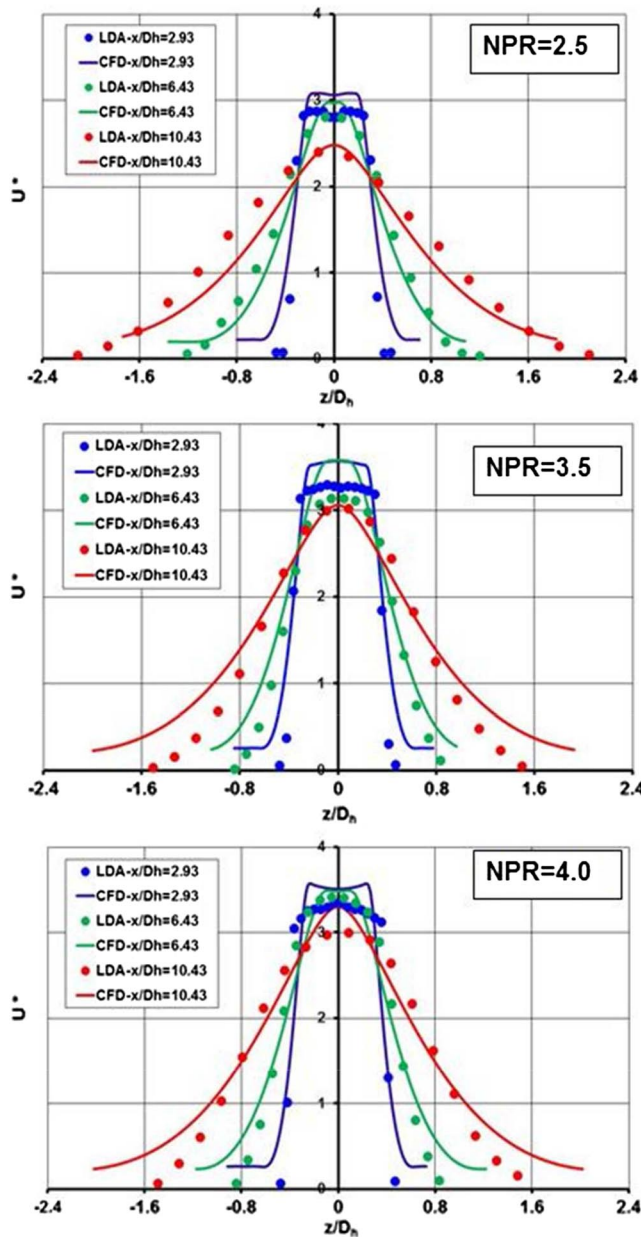


Fig. 6 Mean axial U^* velocity on minor axis, with clean nozzle.

curves in Figs. 4 (U^*) and 5 (u^* and w^*). At the design NPR, only marginal changes are seen, with the small oscillations for the clean nozzle being even smaller in the scarfed case. As will become clear in the following, strong secondary flows associated with scarfing mean the maximum axial velocity location is no longer necessarily uniquely associated with the centerline as in the clean nozzle. These secondary flows may lead to bifurcation of the jet core, and the peak velocity is then no longer on the centerline because unmixed jet core fluid penetrates to a further downstream distance at off-centerline locations. It is thus unwise to define potential core length purely from centerline data for scarfed nozzles. Scarfed nozzle results at the design NPR (Fig. 4; middle) show that the location where centerline velocity begins to decay and the initial decay rate do not seem to have changed (note that the centerline nozzle exit plane is now at $x/D_h = 3.12$ for the scarfed nozzle). Larger differences in centerline characteristics are apparent in Figs. 4 (top) and (bottom) for over- and underexpanded conditions, but examination of spanwise characteristics in the following show that these are caused by secondary flow effects rather than shear layer growth effects. At NPR = 2.5, velocity decay begins much further downstream as compared to the clean nozzle ($10.5D_h$ vs $4.3D_h$), whereas the opposite occurs at an underexpanded

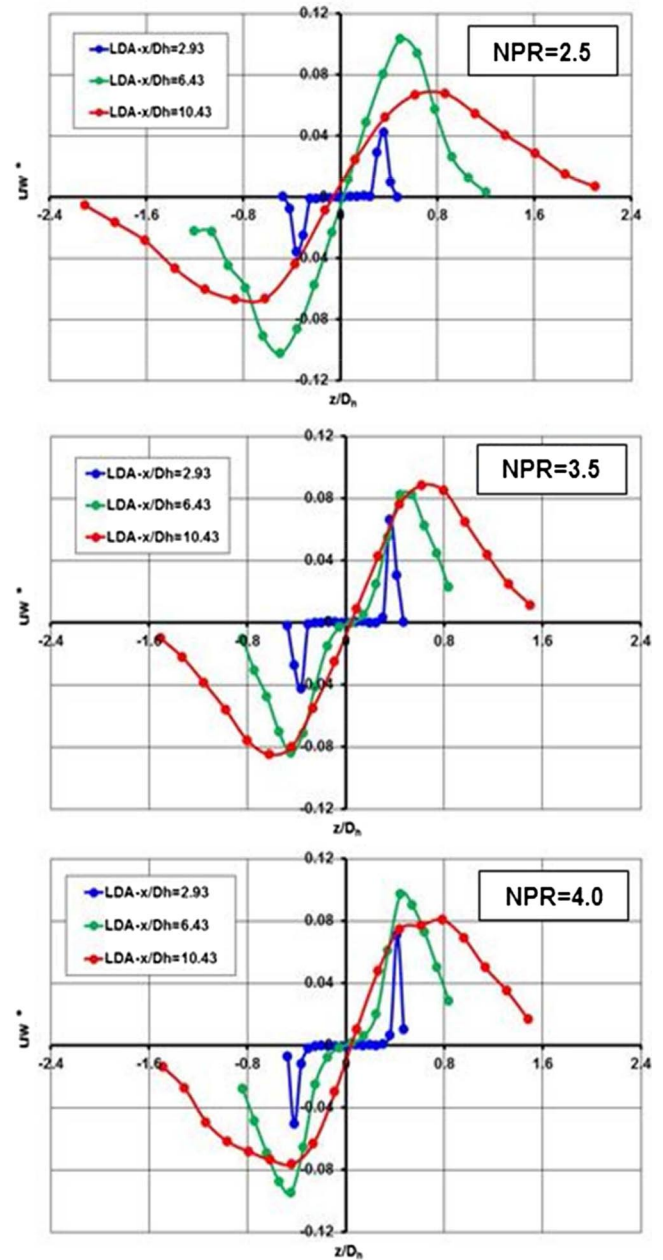


Fig. 7 Reynolds shear stress uw^* on minor axis, with clean nozzle.

NPR ($5.3D_h$ vs $6.5D_h$). The reasons for these offdesign changes are best explained on examination of plume axial cross-section contours, as provided in the following. The initial decay rate has been affected by scarfing, particularly for the underexpanded NPR where a noticeable change is observed between $10 < x/D_h < 20$. The scarf-induced change in centerline behavior is also visible in the development of the normal stresses shown in Fig. 5. At the design NPR, scarfed nozzle data follow development of the clean nozzle u^* stress; for the overexpanded NPR, the increase in turbulence with scarfing is shifted downstream relative to the clean nozzle but upstream for the overexpanded case. It is also noticeable that the relative values of u^* and w^* are as expected from shear-generation considerations for all scarfed nozzle conditions with all shock location “spikes” greatly reduced, indicating screech has been suppressed by scarfing.

A better impression of different clean vs scarfed flow development is gained by examining the minor-axis plane contour plots in Fig. 10. A slightly wider scarfed nozzle plume at NPR = 2.5 but a narrower spread at the other two NPRs may be deduced from contour comparison in the furthest downstream region. The scarfed nozzle pressure adjustment regions extend further downstream for overexpanded flow but less far at the underexpanded NPR.

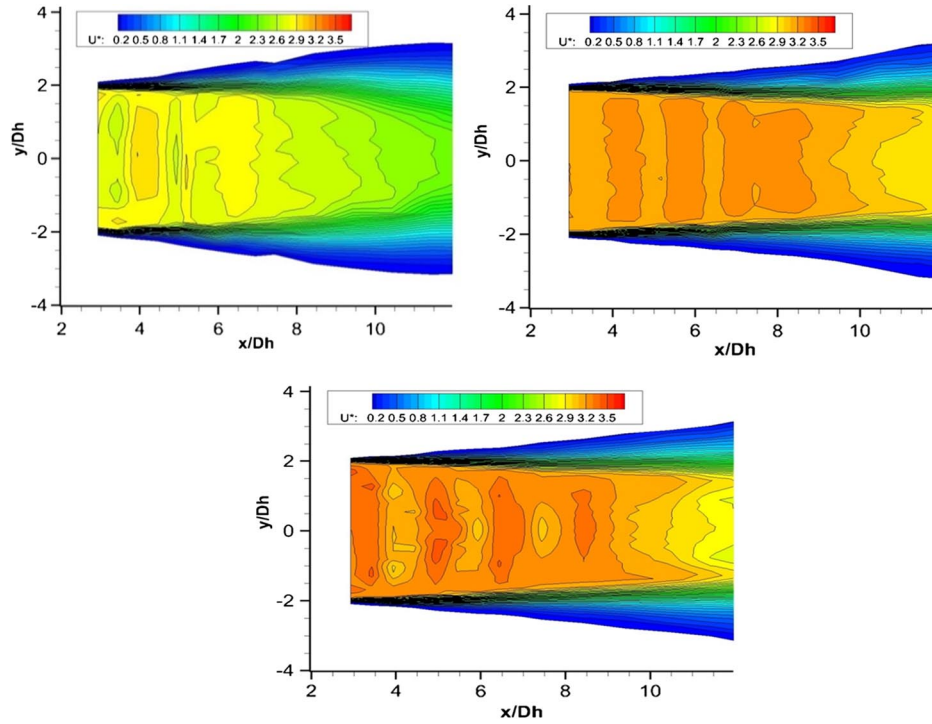


Fig. 8 Mean axial velocity U^* contours on major axis, with clean nozzle: NPR = 2.5 (top left), NPR = 3.5 (top right), and NPR = 4.0 (bottom).

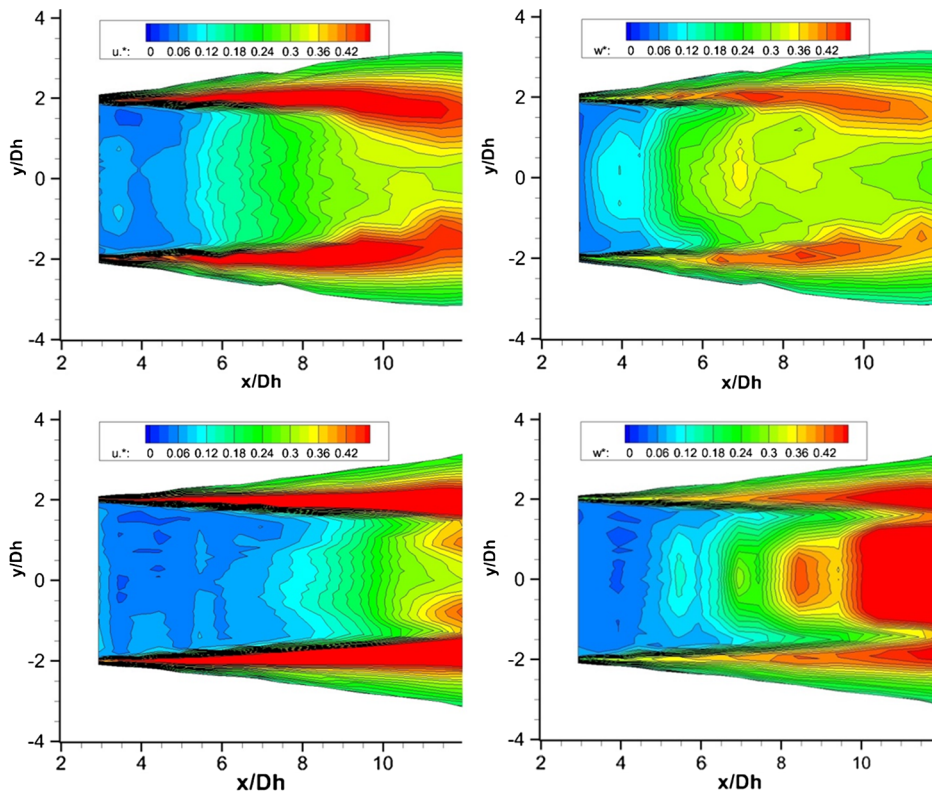


Fig. 9 Axial rms (u^* ; left) and transverse rms (w^* ; right) contours on major axis, with clean nozzle: NPR = 2.5 (top), and NPR = 4.0 (bottom).

The primary cause of different flow development is revealed in the constant- x plume cross sections at the two offdesign NPRs in Fig. 11 (note that CFD results are also shown in this figure and are discussed in the next section). At the station closest to nozzle exit, the rectangular shape of the nozzle exit is maintained for both NPRs, although some indication of scarf-induced secondary flow (y - z plane) effects may just be seen via contour distortions near the symmetry plane (both NPRs) and at the plume corners (NPR = 4.0).

The jet boundary shape in these regions is affected by secondary motions (streamwise vorticity) as well as turbulent diffusion. For NPR = 2.5, these two processes combine to cause radically different spreading rates in the two cross-stream directions. By $x/D_h = 12.0$, the cross-section aspect ratio has changed from 2:1 (half-section) to approximately 1:1, although still not overall axisymmetric. A small circular high-velocity core region is surrounded by four-lobe-shaped contours with maximum penetration on the major/minor-axis

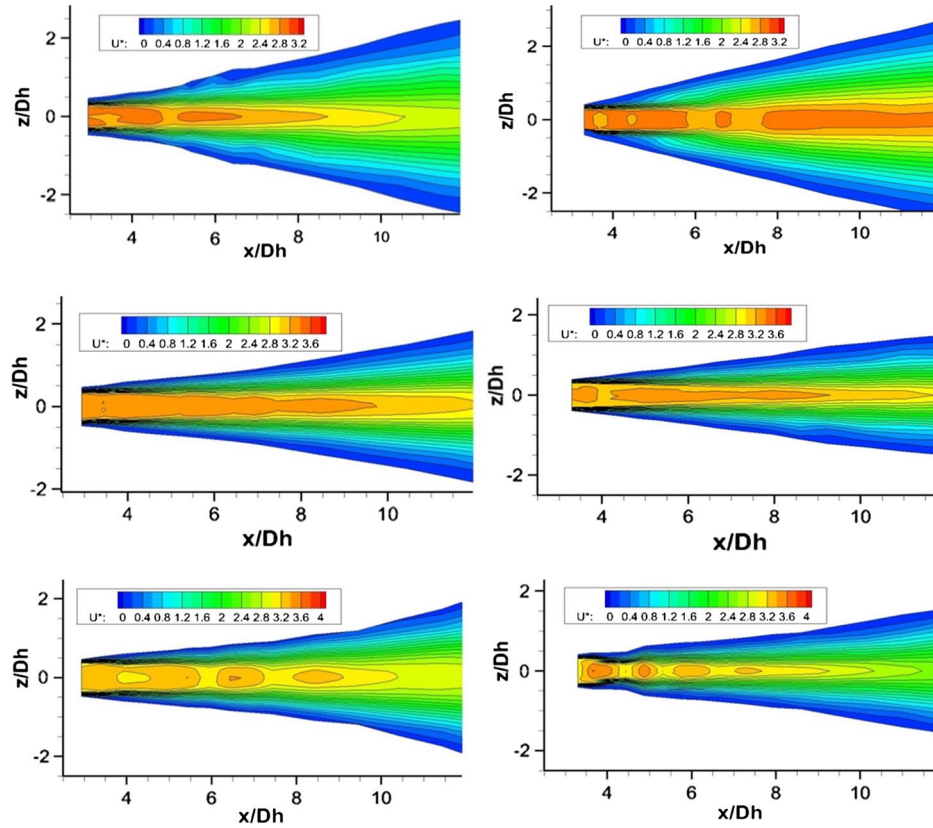


Fig. 10 Contours of axial velocity U^* on plume minor-axis plane clean nozzle (left) vs scarfed nozzle (right): $\text{NPR} = 2.5$ (top), $\text{NPR} = 3.5/3.15$ (middle), and $\text{NPR} = 4.0$ (bottom).

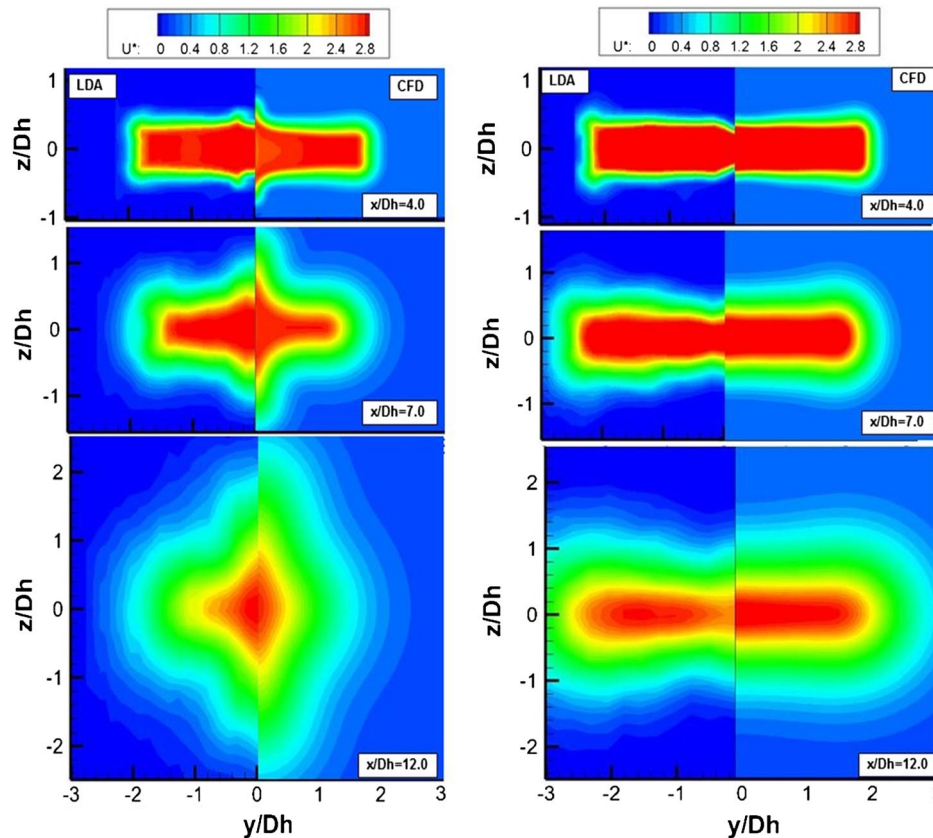


Fig. 11 Contour maps of axial velocity U^* for scarfed nozzle: $\text{NPR} = 2.5$ (left), 4.0 (right) on planes $x/D_h = 4.0$ (top), 7.0 (middle), and 12.0 (bottom).

planes. The plume edge has not penetrated any further in the major-axis direction than its location at nozzle exit, whereas it has moved substantially in the minor-axis direction.

For the underexpanded flow, the interplay between mixing and secondary velocity convection produces a completely contrasting result: the plume retains its rectangular-like shape for the whole

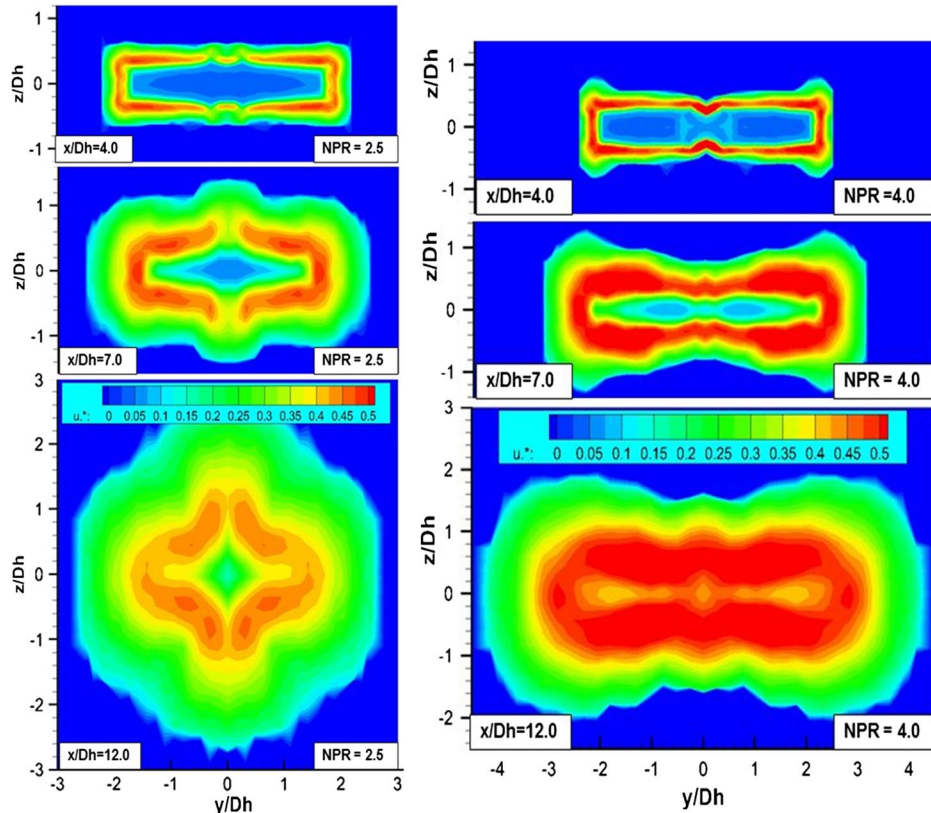


Fig. 12 Contour maps of axial u^* rms velocity, with scarfed nozzle for NPR = 2.5 (left) and 4.0 (right) on planes $x/D_h = 4.0$ (top), 7.0 (middle), and 12.0 (bottom).

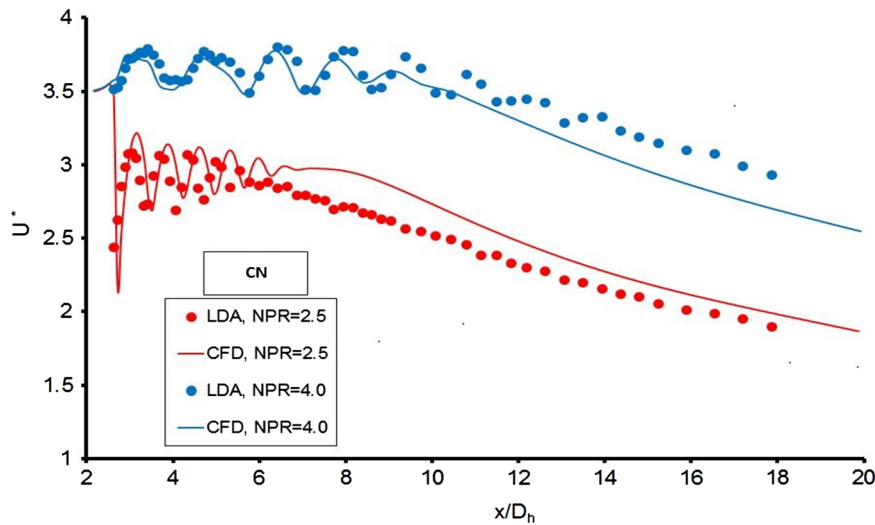


Fig. 13 Comparison of LDA and CFD for centerline axial velocity for clean nozzle.

downstream distance. Plume growth occurs in both cross-stream directions, but at roughly similar rates to maintain the initial shape, with turbulent diffusion leading to rounded corners. In addition, a bifurcation appears with two peaks of velocity observed on either side of the centerline at $x/D_h = 12.0$.

Finally, Fig. 12 shows that the development of the u^* normal stress rms provides a similar picture of the distorting effect of the secondary motions. It is noticeable that the peak amplitude of u^* has reached a maximum at $x/D_h = 7.0$ and then begins to decay for NPR = 2.5; whereas for NPR = 4.0, the peak value is maintained over the whole downstream length, with the high-turbulence region thickening substantially. The explanation for these different behaviors is that the movement of the plume boundary due to secondary motion and

turbulent spreading is coincident and reinforcing in the case of NPR = 2.5 but contrary and cancelling for NPR = 4.0.

C. CFD Results

CFD predictions of jet plumes from clean and scarfed nozzles are now presented, with emphasis given to offdesign NPRs because measurements showed this is where the largest effects of scarfing were observed. Figure 13 shows a comparison of clean nozzle LDA measurements and CFD predictions along the plume centerline. The agreement is generally reasonable. The appearance of a sharp drop in velocity at the nozzle exit to (just) subsonic levels is captured well in the overexpanded case. The number and amplitude of the shock cells

are well predicted, although their location is shifted downstream for the overexpanded NPR. The potential core length is overpredicted (as expected for an eddy-viscosity-based RANS turbulence model), particularly for $\text{NPR} = 2.5$. However, the experimentally observed tendency for the potential core length to increase between over- and underexpanded conditions is captured in the CFD results. The initial velocity decay rate is not changed appreciably by the NPR and is close to that observed in experiments. However, the larger potential core error for $\text{NPR} = 2.5$ causes downstream plume velocities to be overpredicted, but they are underpredicted for $\text{NPR} = 4.0$. Finally, to examine jet profile axial development, measurements and predictions on the minor-axis plane are compared in Fig. 4 (similar characteristics are seen for major-axis plane development, and thus not shown here). The agreement is again reasonable, with a slightly too slow spreading rate for $\text{NPR} = 2.5$ and a too rapid rate for $\text{NPR} = 4.0$, which is consistent with the centerline results.

Predictions for the scarfed case are compared with measurements for the two offdesign NPRs in Fig. 11, focusing on the plume cross-sectional shape change seen via axial velocity contour maps in three constant- x planes. The measured trend is captured well by CFD, in that a four-lobed shape is predicted to appear at overexpanded conditions; whereas a rectangular shape is maintained in underexpanded flow. A more detailed assessment leads to the conclusion that the secondary velocities that control the shape change are overpredicted at overexpanded conditions and underpredicted for underexpanded flow. Although CFD correctly predicts a four-lobed shape for $\text{NPR} = 2.5$, predicted secondary velocities are clearly too strong because CFD shows an overshoot compared to measurements at $x/D_h = 12.0$, with the z -direction size of the predicted plume incorrectly exceeding the y -direction size. The underexpanded case is also not exactly

reproduced in CFD results because no evidence of bifurcation is shown in predicted velocity contours, unlike in the measured data. There was no evidence at further downstream locations of any predicted bifurcation, so it seems likely that the secondary velocities that cause the peak velocity to move off centerline are underpredicted (see the following). In addition to these global discrepancies, a local error appears with respect to the secondary velocities induced at the sharp central corner of the scarfed geometry. The contour distortions at the first measurement station are different in the experiments and CFD. For overexpanded flow, two outward-pointing peaks appear on either side of the centerline in the experiments, whereas only a single peak on the centerline is seen in the CFD. Similarly, for underexpanded flow, an inward-pointing distortion appears in measurements but is not seen in the CFD. These comparisons show that, although the current RANS closure captures the trend of scarf effects on plume development at offdesign NPRs qualitatively correctly, there is still a lack of predictive capability in terms of the correct amplitude and local details of the scarf-induced secondary flows.

In spite of the differences noted previously between the predictions and measurements, CFD can still be usefully exploited to clarify the interacting mechanisms that cause scarfing to produce such different effects at over- and underexpanded conditions. Crossplane secondary flows are driven by both static pressure gradients and turbulent mixing. For the clean geometry, the nozzle exit static pressure is broadly uniform, although only at the design NPR is this at ambient pressure. Examination of the CFD-predicted clean nozzle exit static pressure at $\text{NPR} = 3.5$ indicates that the central 90% of the nozzle exit area was at $\sim 94\% p_{\text{ambient}}$. Thin regions near the walls showed slightly lower pressure ($\sim 91\% p_{\text{ambient}}$) but, in general, the close-to-ambient condition was observed. For $\text{NPR} = 2.5$, the nozzle exit

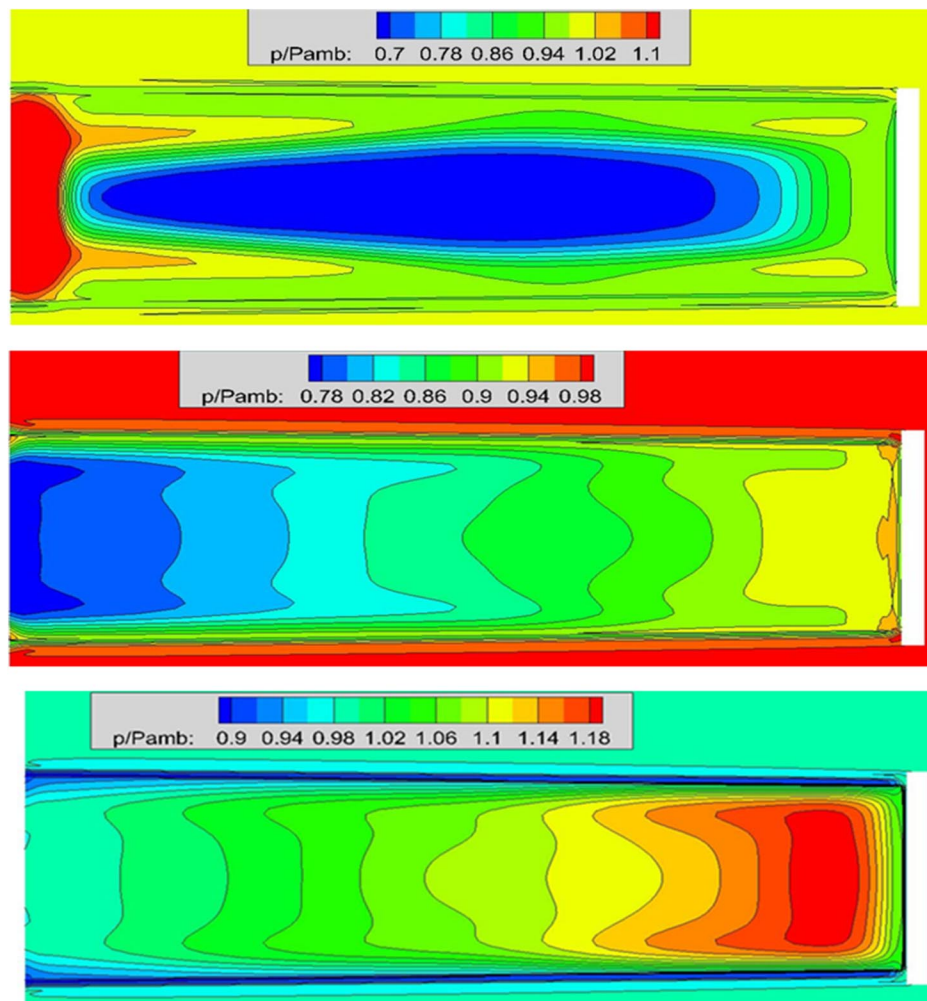


Fig. 14 CFD prediction of static pressure field on scarfed nozzle exit plane: $\text{NPR} = 2.5$ (top), $\text{NPR} = 3.15$ (middle), and $\text{NPR} = 4.0$ (bottom).

static pressure (again uniform over $\sim 90\%$ of the area) was $\sim 66\% p_{\text{ambient}}$; whereas at $\text{NPR} = 4.0$, it was $\sim 107\% p_{\text{ambient}}$.

Scarfed predictions for the nozzle exit plane static pressure are presented in Fig. 14. Note that what is shown is the static pressure over the angled nozzle exit. Thus, the plane shown in Fig. 14 is canted backward at 35° to the x axis, with the nozzle centerline on the left (axial location: $x/D_h = 3.12$) and the nozzle sidewall (white rectangle) on the right (axial location: $x/D_h = 1.72$). The angled exit area means that pressure exchange between the internal nozzle fluid and the ambient fluid is able to occur over the entire axial distance between the nozzle exit at the sidewall and at the centerline nozzle tip. In addition, because different expansion lengths are experienced by fluid on the centerline and fluid at the wall, the static pressure at these two locations will also clearly be different.

The pressure difference between the internal nozzle fluid and the ambient will of course drive crossplane secondary flow. The predicted static pressure fields in Fig. 14 indicate substantial pressure differences exist over the scarfed nozzle exit plane. At the design NPR, the pressure is generally lower than ambient, particularly at the centerline. At the overexpanded NPR, the pressure is again less than ambient, but the pressure difference is now substantially larger. At the underexpanded NPR, the pressure difference is predominantly positive, especially at the nozzle sidewall. These pressure differences will, in particular, drive flow in the spanwise negative y direction at design and overexpanded NPRs but in the positive y direction at the underexpanded NPR. Pressure-driven secondary flows will be superimposed on those generated by the turbulent entrainment process, which in general will be directed from the ambient in toward the jet core. Figure 15 illustrates the predicted secondary flow patterns (vectors formed from y/z components only) at $x/D_h = 7.0$ for the two offdesign NPRs for the clean nozzle (top) and the scarfed geometry (bottom).

Clean nozzle results illustrate the expected vortex ring-driven secondary flow pattern for a rectangular jet, which is (strongly) inward toward the nozzle center on the major axis and (weakly) outward from the jet core toward the jet edge on the minor axis.

Superimposed on this is the overall turbulent diffusion-driven entrainment velocity (radially inward from ambient toward the jet center). These mechanisms combine to form the secondary flow pattern seen in Fig. 15 (top), which is generally similar at over- and underexpanded conditions.

Figure 15 (bottom) illustrates how this pattern changes due to scarfing at the overexpanded NPR (left) and underexpanded NPR (right). In addition to the two mechanisms described previously, a third mechanism is the pressure deficit (overexpanded) or excess (underexpanded) between the jet core and ambient. For $\text{NPR} = 2.5$, this drives flow from the ambient into the jet core in the major-axis direction; for $\text{NPR} = 4.0$, the reverse occurs. This third mechanism reinforces the vortex ring-induced secondary velocity for $\text{NPR} = 2.5$ but opposes it for $\text{NPR} = 4.0$. The consequence is that the tendency for the secondary flows to change the rectangular shape of the jet is greatly enhanced for $\text{NPR} = 2.5$ but weakened for $\text{NPR} = 4.0$. The flow patterns shown in Fig. 15 explain the predicted cross-sectional shapes seen in Fig. 11. For $\text{NPR} = 2.5$, the interaction between minor-axis outflow and inward entrainment flow creates a clockwise-rotating vortex near the central symmetry plane in the upper half of the jet (and a reverse vortex in the lower half). This vortex is clearly responsible for the two lobes appearing in the z direction in the upper/lower halves of the jet. The opposite is observed for $\text{NPR} = 4.0$, where vortex ring-induced flows and scarf-induced flows oppose each other and almost cancel out. The interaction/impingement of these two flow mechanisms now creates just a small counterclockwise vortex in the upper jet half at approximately two-thirds of the jet width, and the contours in Fig. 11 retain their rectangular shape. The outward-directed secondary flow on the major axis seen in Fig. 15 (bottom right) would result in the plume bifurcation seen in the scarfed experiments at the underexpanded NPR, but it is clearly too weak to cause this to occur in the CFD predictions. However, it may be that, because grid independence checks were carried out for the clean nozzle only, insufficient resolution of the sharp central and nozzle edge corners in the scarfed

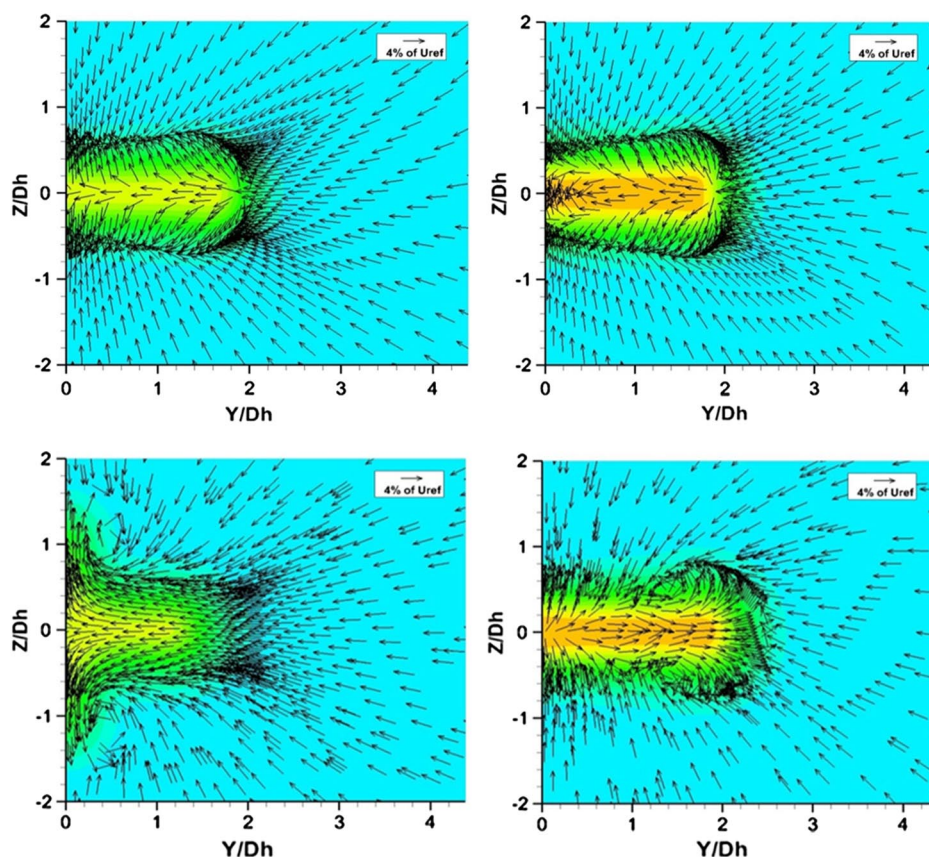


Fig. 15 CFD prediction of secondary velocity vectors on plane $x/D_h = 7$: clean nozzle (top), scarfed nozzle (bottom), $\text{NPR} = 2.5$ (left), and $\text{NPR} = 4.0$ (right).

geometry may have contributed to this underprediction, and further grid refinement studies would be useful.

VI. Conclusions

Two rectangular convergent-divergent nozzles with clean and scarfed exit planes were designed. Schlieren visualization and LDA measurements were carried out to capture plume development in the near field at overexpanded, (close to) design, and underexpanded conditions. RANS CFD studies using the Spalart-Allmaras turbulence model were also completed for both clean and scarfed nozzle geometries at the same operating conditions.

Schlieren images for the clean nozzle showed the expected differences with NPR, with overexpanded flow producing 2-D transverse pressure-wave structures, and underexpanded conditions displaying stronger three-dimensionality. Some evidence of screech-induced jet flapping in the minor direction was observed, and the NPRs selected for detailed measurements were chosen to avoid the region where screech was strong enough to influence mean flow development. The effect of scarfing on these patterns was to reduce the design NPR, to produce generally increased 3-D wave patterns, and to suppress any occurrence of screech.

Detailed LDA measurements were provided along the centerline and for profiles on major/minor axes. For the clean nozzle, mild screech effects were observed via the excess of transverse over axial velocity fluctuations on the jet centerline. However, in the shear layers where momentum mixing dominated the jet development, there was no evidence of significant screech effects. Measurements also showed that the peak nondimensional axial velocity at the nozzle exit increased with the NPR, indicating small changes in the nozzle discharge coefficient. Saddle-shaped velocity profiles were noted in the downstream region for the underexpanded case, but they were insufficient to cause axis switching. Scarfing produced a significant effect on jet centerline development; the location where velocity decay began was moved downstream for overexpanded flow but shifted upstream slightly at the underexpanded NPR. Data on the plume cross section revealed the strongest effect of scarfing was at offdesign conditions, particularly for overexpanded flow where a large plume cross-section distortion into a four-lobed shape was observed.

For a clean nozzle, the RANS CFD predicted flow development reasonably accurately, although the expected overprediction of potential core length was observed. The effect of the NPR on the developing plume cross section for the scarfed nozzle at offdesign conditions was reproduced, but only qualitatively. At the overexpanded condition a four-lobed shape was predicted but the strength of secondary velocities driving this was overestimated. Examination of the predicted scarfed nozzle exit static pressure explained the very different behavior at over- and underexpanded conditions. For the former, the pressure-driven secondary velocities reinforced the secondary velocities that drove the jet cross-section shape change, whereas they reduced these at the underexpanded NPR. Given that the amplitude of the secondary velocities was not captured accurately by the RANS CFD applied here, further grid refinement studies are recommended for the scarfed nozzle case, and the present experimental data should be further explored using alternative RANS or LES turbulence closures to identify the optimum CFD approach for prediction of scarfed nozzle effects.

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